UNIT - I

Measure of central tendency:

A measure of central tendency of a single value with in the range of the ~~marks~~ entire marks of the data used to reference the whole data.

The measures of central tendency is also called measures of averages.

Types of Averages:

There are two types of Averages

(1) Mathematical Averages.

(2) Positional Averages.

(1) Mathematical Averages:

There are three types in Mathematical Averages - 1. Arithematic Mean (A.M) (or) Mean (or) Average

2. Geometric Mean (G.M)

3. Harmonic Mean (H.M)

2) Positional Averages:

There are two types in positional Averages.

1. Median.
2. Mode.

Arithematic Mean:

Arithmatic Mean (or) Average (or) Mean all are same.

Average mean sum of observations divided by total number of observations.

S.No	Types of Series	Mean	Identification
1	Individual series	Mean $= \bar{x} = \frac{\sum x_i}{n}$, where n = Total number of observations	$x_1, x_2, x_3 \text{-----} x_n$
2.	Discrete series.	(a) <u>Direct method</u>: Mean $= \bar{x} = \frac{\sum fx}{N}$, where N = sum of frequencies $= \sum f$ (b) <u>Short-cut method</u>: Mean $= \bar{x} = A + \frac{\sum fd}{N}$ where A = Arbitrary point (Assumed Mean) It is value of x, in middle of distribution. $d = x - A$, $N = \sum f$	$x : x_1 \mid x_2 \mid \text{---} \mid x_n$ $f : f_1 \mid f_2 \mid \text{---} \mid f_n$
3.	Continuous series	(a) <u>Direct Method</u>: Mean $= \bar{x} = \frac{\sum fx}{N}$ where x = mid value of each class (b) <u>Short-cut Method</u> { <u>step deviation Method</u>): Mean $= \bar{x} = A + \left(\frac{\sum fd}{N}\right) \times c$ x = mid values of each class A = Arbitrary point. $N = \sum f$ $d = \frac{x - A}{c}$ c = common factor (or) common magnitude (or) of class interval.	C.I: $x_0 - x_1 \mid x_1 - x_2 \mid \text{----} \mid x_{n-1} - x_n$ f: $f_1 \mid f_2 \mid \text{---} \mid f_n$

Problems:

1. Calculate the mean value to the following value, 43, 48, 65, 57, 31, 60, 37, 48, 78, 59, and 45

A. Mean $= \bar{x} = \frac{\sum x_i}{n}$, where $n =$ Total number of observations

$$= \frac{43+48+65+57+31+60+37+48+78+59+45}{11}$$

$$= \frac{571}{11}$$

$= 51.9091$ <∵ Individual series.

$$\boxed{\bar{x} = 51.9091}$$

Q. Find the Arithmetic mean of the following frequency distribution.

x :	1	2	3	4	5	6	7
f :	5	9	12	17	14	10	6

A. Given frequency distribution belongs to discrete series.

$$\text{Mean} = \bar{x} = \frac{\sum fx}{N} \quad (1), \text{ where } N = \sum f$$

x	f	fx
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
	$N = 73$	$\sum fx = 299$

Substitute all these values in eq (1), we get.

$$\text{Mean} = \bar{x} = \frac{299}{73} = 4.09$$

3. Calculate the Arithmetic mean of the marks from the following table.

Marks :	0–10	10–20	20–30	30–40	40–50	50–60
No. of students :	12	18	27	20	17	6

A. Given frequency distribution belongs to continuous series

Mean $= \bar{x} = \frac{\Sigma fx}{N} \rightarrow (1)$, x = mid value of each class

$N = \Sigma f$

Marks (C.I)	No: of Students (f)	Mid value (x)	fx
0 – 10	12	$\frac{0+10}{2} = 5$	60
10 – 20	18	$\frac{10+20}{2} = \frac{30}{2} = 15$	270
20 – 30	27	25	675
30 – 40	20	35	700

40-50	17	45	765
50-60	6	55	330
	N = 100		$\Sigma fx = 2800$

Put all these values in eq (1), we get.

$$\text{Mean} = \bar{x} = \frac{2800}{100} = 28$$

$$\therefore \boxed{\bar{x} = 28}$$

Merits and Demerits of Arithmetic mean:

Merits:

1. It is easy to understand and easy to calculate.
2. It is based upon all observations.
3. It is rigidly defined.
4. It is least effected by fluctuations of sampling (i.e., A.M is a stable average)

Demerits:

1. It cannot be determined by inspection.
2. A.M cannot be used if we are dealing with qualitative characteristics which cannot be measured quantitatively.
   Ex: Intelligence, Honesty, Beauty
3. ~~Athm~~ A.M cannot be obtained if single observation is missing.
4. A.M is effected ~~to~~ very much by extreme values.

Median :

Median is a value which divides the given data into two equal parts. that can be arranged in an order either ascending or descending.

Median - individual series :

1. Arrange the observations in ascending or descending order.

$$\text{Median} = \frac{n+1}{2}\text{ item}, \text{ if } n \text{ is } \cancel{\text{even}}\ \text{odd}.$$

$$= \frac{\left[\frac{n}{2}\text{ item} + \left(\frac{n}{2}+1\right)\text{ item}\right]}{2} \quad \text{if } n \text{ is } \cancel{\text{odd}}\ \text{Even}$$

where n = Total number of items (observations).

1. Oblain the value of median from the following data.

120, 170, 100, 110, 180, 220, 160

A. Ascending order of given data

100, 110, 120, 160, 170, 180, 220

n = No: of observations = 7 (odd)

Median = $\frac{n+1}{2}$ item

$= \frac{7+1}{2} = 4$ item

$= 160$

[or]

Median = Middle value = 160

Median- discrete series :

In case of discrete series the following steps are taken for calculating the median.

1. Arrange the data in Ascending order if not arranged.
2. Obtain the total frequency $N = \Sigma f$.
3. Obtain the cummulative frequency (c.f).
4. Find $\frac{N}{2}$.
5. See c.f just greater than $\frac{N}{2}$
6. The corresponding x value is the Median.

1. obtain the median for the following frequency distribution.

x :	1	2	3	4	5	6	7	8	9
f :	8	10	11	16	20	25	15	9	6

A-

	x	f	c.f
	1	8	8
	2	10	8+10 = 18
	3	11	18+11 = 29
	4	16	29+16 = 45
M	5	20	45+20 = 65 c.f
	6	25	65+25 = 90
	7	15	90+15 = 105
	8	9	105+9 = 114
	9	6	114+6 = 120
		N = 120	

Now, $\frac{N}{2} = \frac{120}{2} = 60$, c.f = 65 and the corresponding x value is 5.

$\therefore$ Median = M = 5

Median - continuous series:

$$\text{median} = L + \left(\frac{\frac{N}{2} - c.f}{f} \right) \times c$$

where, L = lower limit of median class

$N = \Sigma f$

f = frequency of median class.

$c.f$ = cummulative frequency of class preceding the median class.

c = Width of median class.

* First find $\frac{N}{2}$

1. The following table gives the weekly wages in Rs in a certain commertial organisation.

weekly wages :	30-32	32-34	34-36	36-38	38-40	40-42
f :	2	9	25	30	49	62

weekly wages :	42-44	44-46	46-48	48-50.
f :	39	20	11	3

Ans:-

C.I	f	c.f
30-32	2	2
32-34	9	9+2=11
34-36	25	25+11=36
36-38	30	66
38-40	49	115 (c.f
40-42 (L)	62 (f)	177
42-44	39	216
44-46	20	236
46-48	11	247
48-50	3	250
	N=250	

Now, $\frac{N}{2} = \frac{250}{2} = 125$

$\Rightarrow L = 40, \quad f = 62, \quad c.f = 115, \quad c = 2.$

$\therefore$ Median $= L + \left(\frac{\frac{N}{2} - c.f}{f}\right) \times c$

$= 40 + \left(\frac{125 - 115}{62}\right) \times 2$

$= 40 + 0.1612 \times 2$

$= 40 + 0.3225$

$= 40.3226$

Merits and Demerits of median:

Merits:

1. It is rigidly defined.
2. It is easy to understand and easy to calculate
3. It is not effected by extreme values.
4. It can be calculated for distributions with open-end classes.

Demerits:

1. In case of even no: of observation median cannot determined exactly.
2. It is not based on all observations.
3. As compared with mean it is effected much by fluctuations of sampling.

Uses:

1. It is used in further statistical Analysis.
   Ex: Mean-deviation, co-efficient of skewness.
2. It is used to find typical values in problems concerning wages, distribution of welth etc.

Mode :

Mode is a value which is repeated maximum number of times.

Mode - individual series :

1. calculate mode from the values 15, 32, 29, 17, 32, 25, 32, 29, 20, 35, 25

A. Here the value 32 is repeated three times (max)

∴ Mode = 32.

2. Calculate mode for the values 12, 13, 17, 18, 19, 19, 20, 21, 21, 22, 24, 27, 19, 31, 31, 31, 32, 31

A. Here the value 31 is repeated maximum times (four)

∴ Mode = 31.

Mode - discrete series:

In discrete series the mode value is the value of 'x' corresponding to maximum frequency

In some cases there may be two or more highest frequencies, in such cases mode cannot be defined exactly. we use bi-modal (or) Multi-modal.

i.e., $\boxed{\text{Mode} = 3 \times \text{median} - 2 \times \text{mean}}$

1. Find the mode from the following data.

x :	20	30	40	50	60	70
f :	5	20	25	25	12	10

A. Here the highest frequency is 25. and it is repeated twice. So, the given problem has bi-modal.

$$\text{Mode} = 3\,\text{median} - 2\,\text{mean}.$$

x	f	c.f	fx
20	5	5	100
30	20	25	600
40 (M)	25	50 (c.f)	1000
50	25	75	1250
60	12	87	720
70	10	97	700
	$N = 97$		$\Sigma fx = 4370$

$\therefore$ Mode $= 3 \times 40 - 2 \times 45.0515$

$= 120 - 90.1031$

$\boxed{\text{Mode} = 29.897}$

(i) Mean $= \frac{\Sigma fx}{N} = \frac{4370}{97} = 45.0515$

(ii) Median: $\frac{N}{2} = \frac{97}{2} = 48.5$, $\therefore$ Median $= 40$.

Mode - continuous series:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times c$$

where, L = lower limit of modal class.

f_1 = frequency of the modal class.

f_0 = frequency of the class preceding the modal class.

f_2 = frequency of the class succeeding the modal class.

c = common factor (or) common magnitude of modal class.

1. Find the mode for the following distribution.

C.I	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
f	5	8	7	12	28	20		

A. Here the maximum frequency is 28. Thus the modal class is 40 – 50.

$\therefore$ $L = 40$, $f_1 = 28$, $f_0 = 12$, $f_2 = 20$, $c = 10$.

$$\therefore \text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times c$$

$$= 40 + \left(\frac{28 - 12}{2(28) - 12 - 20}\right) \times 10$$

$$= 40 + \left(\frac{16}{24}\right) \times 10 = 46.6667$$

Merits and Demerits of Mode :

Merits :

1. It is easy to calculate and easy to understand.
2. It can be found by inspection.
3. It is not affected by extreme values as in the average.
4. The value of mode can be determined by the graphic method

Demerits :

1. Mode is ill-defined.
2. It is not based on all observations.
3. It is not capable of further mathematical treatement.
4. As compared with mean, mode is affected to greater extent by the fluctuation of sampling.

Uses of mode:

1. When a quick and approximate measures of central tendency desired.
2. To find the ideal size.

Ex:- In bussiness forecasting, in manufacturing of readymade garments, shoes etc.

Geometric mean: (GM)

Geometric mean of 'n' numbers is defined as the n^{th} root of product of 'n' numbers.

i.e., $G.M = \sqrt[n]{x_1 \times x_2 \times \cdots \times x_n}$

Types of Series	G.M.
Individual series	$G.M = \text{Antilog}\left(\frac{\Sigma \log x}{n}\right)$
Discrete series	$G.M = \text{Antilog}\left(\frac{\Sigma f.\log x}{N}\right)$. where $N = \Sigma f$
continuous series	$G.M = \text{Antilog}\left(\frac{\Sigma f.\log x}{N}\right)$ where, x = middle value of each class $N = \Sigma f$.

1. Calculate G.M for the values 2574, 475, 5, 0.8, 0.005, 0.009.

A. The problem belongs to individual series.

x	$\log x$
2574	3.4106
475	2.6767
5	0.6990
0.8	−0.0969
0.005	−2.3010
0.009	−2.0458
	$\Sigma \log x = 2.3426$

n = number of observations = 6

$$G.M = \text{Antilog}\left(\frac{\Sigma \log x}{n}\right)$$

$$= \text{Antilog}\left(\frac{2.3426}{6}\right)$$

$$= \text{Antilog}\,(0.3904)$$

$$= 2.4570$$

2. Calculate G.M for following data.

C.I :	100-110	110-120	120-130	130-140	140-150	150-160
f :	12	18	25	22	18	10

A.

C.I	f	Mid values (x)	log x	f. log x
100-110	12	$\frac{100+110}{2} = 105$	2.0212	24.2543
110-120	18	115	2.0607	37.0926
120-130	25	125	2.0969	52.4228
130-140	22	135	2.1303	46.8673
140-150	18	145	2.1614	38.9046
150-160	10	155	2.1903	21.9033
	N = 105			$\Sigma f \cdot \log x = 221.4449$

$$\therefore G.M = \text{Antilog}\left(\frac{\Sigma f.\log x}{N}\right)$$

$$= \text{Antilog}\left(\frac{221.4449}{105}\right)$$

$$= \text{Antilog}\ (2.1090)$$

$$= 10^{2.1090}$$

$$= 128.5287.$$

Harmonic mean : (H.M)

Harmonic mean is based on the reciprocals of "numbers averaged".

S.NO	Type of series	H.M.
1.	Individual series	$H.M = \frac{n}{\Sigma\left(\frac{1}{x}\right)}$
2.	Descrete series	$H.M = \frac{N}{\Sigma\left(\frac{f}{x}\right)}$
3.	continuous series	$H.M = \frac{N}{\Sigma\left(\frac{f}{x}\right)}$ where, x = mid values.

1. Calculate H.M for 2574, 475, 5, 0.8, 0.08, 0.005, 0.009

A. Given series is individual series and

n = No: of observations = 7

x	$\frac{1}{x}$
2574	1/2574 = 0.004
475	0.0021
5	0.2
0.8	1.25
0.08	12.5
0.005	200
0.009	111.111
	$\Sigma fx = 325.0636$

$$\therefore H.M = \frac{n}{\Sigma\left(\frac{1}{x}\right)} = \frac{7}{325.0636}$$

$$= 0.0215$$

Q. Calculate H.M from the following data.

x :	10	20	30	40	50	60
f :	5	16	18	12	24	15

A. The given distribution is discrete series.

$$\therefore H.M = \frac{N}{\Sigma\left(\frac{f}{x}\right)} \longrightarrow ①$$

x	f	$\frac{f}{x}$
10	5	0.5
20	16	0.8
30	18	0.6
40	12	0.3
50	24	0.48
60	15	0.25
	$N=90$	$\Sigma\left(\frac{f}{x}\right) = 2.93$

eq ① becomes $\Rightarrow$ $H.M = \frac{90}{2.93}$

$= 30.7167$

3. calculate H.M of the following data.

Marks :	30-40	40-50	50-60	60-70	70-80	80-90	90-100
frequency :	15	13	8	6	15	7	6

A. The given distribution is continuous series.

$\therefore H.M = \frac{N}{\Sigma\left(\frac{f}{x}\right)} \rightarrow (1)$, where x = Mid value of each class

C.I	f	x (Mid)	f/x
30-40	15	35	0.4286
40-50	13	45	0.2889
50-60	8	55	0.1455
60-70	6	65	0.0923
70-80	15	75	0.2
80-90	7	85	0.0824
90-100	6 ; N=70	95	0.06321

$\rightarrow \Sigma\left(\frac{f}{x}\right) = 1.3009$

eq (1) $\Rightarrow H.M = \frac{70}{1.3009}$

$= 53.8089$

Measures of dispersion :

Dispersion is a measure of variation of items. It measures the extent to which the items vary from some central value. Measures of dispersion shows the average of the differences of various items from an average. For this reason theyre are also called the averages of the second order.

Definition :

→ Dispersion is the measure of the variation of the items.

→ Dispersion is a measure of extent to which the individual items may differ or vary

Purpose of measures of dispersion:

The following are the main purposes of measure of dispersion.

1. To test the reliability of an average.
2. To serve as a ~~blasses of~~ basis for controllar variability
3. To compare two or more series with regard to their variability.
4. To facilitate as a basis for further statistical analysis.

Methods of measuring dispersion:

1. ~~Range~~ :

1. Range
2. Quartile deviation (Q.D)
3. Mean deviation (M.D)
4. standard deviation (S.D)
5. co-efficient variation (C.V)

1. Range:

Range is the difference between highest value and lowest value.

Range = large value - small value

$= L - S$

co-efficient of range $= \frac{L-S}{L+S}$

1. Find the range of values 120, 170, 240, 100, 105, 205, 300, 160, 150, 180. And also calculate its co-efficient.

A. $S = 100$, $L = 300$.

Range $= L - S$

$= 300 - 100 = 200$

co-efficent of range $= \frac{L-S}{L+S}$

$= \frac{200}{400}$

$= 0.5$

Merits:

1. It is simple to calculate and understand.
2. It gives a rough answer.

Demerits:

1. It is not reliable because it is effected by extreme values.
2. It cannot be applied to open-end classes
3. It is not suitable for mathematical treatement.

Uses:

1. Range is used in industry for the statistical quality control of the manufacture product by the construction of control charge.
2. It is useful in studying the variations in the prizes of stock, shares.

2. Quartile deviation (Q.D):

Quartiles are the points which divide the array in four equal parts.

Half of the difference between third and first quartiles is called the quartile deviation (or) semi-inter quartile range.

Quartile deviation - individual series:

$$\text{Quartile deviation} = \frac{Q_3 - Q_1}{2}$$

Where, Q_1 = first quartile (or) lower quartile = the value of $\left(\frac{n+1}{4}\right)$ item.

Q_3 = third quartile (or) upper quartile = the value of $\left[3\left(\frac{n+1}{4}\right)\right]$ item.

$$\text{co-efficient of quartile deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

1. calculate quartile deviation and its co-efficient from the follaoing values.

25, 33, 45, 17, 35, 20, 55

A. Ascending order of given values is.

17, 20, 25, 33, 35, 45, 55

n = no: of values = 7

Q_1 = The value of $\left(\frac{n+1}{4}\right)$ item

= The value of $\left(\frac{7+1}{4}\right)$ item

= The value of 2^{nd} item = 20

Q_3 = The value of $\left[3\left(\frac{n+1}{4}\right)\right]$ item

= The value of 6^{th} item = 45

∴ Quartile deviation Q.D = $\frac{Q_3 - Q_1}{2}$

$= \frac{45-20}{2} = 12.5$

Co-efficient of Quartile deviation = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

$= \frac{45-20}{45+20} = \frac{25}{65}$

$= 0.3846.$

Quartile deviation - discrete series :-

Quartile deviation $Q.D = \frac{Q_3 - Q_1}{2}$

Co-efficient Quartile deviation $= \frac{Q_3 - Q_1}{Q_3 + Q_1}$

Where first write cummulative frequency and then find Q_1 = The size of $\left(\frac{N+1}{4}\right)$ item.

Q_3 = The size of $\left[3\left(\frac{N+1}{4}\right)\right]$ item.

1. Calculate Q.D. and its co-efficient from the following

x :	58	59	60	61	62	63	64	65	66
f :	15	20	32	35	33	22	20	10	8

Ans :-

x	f	c.f
58	15	15
59	20	20+15=35
60 (Q_1)	32	32+35=67 c.f for Q_1
61	35	35+67=102
62	33	33+102=135
63 (Q_3)	22	135+22=157 c.f for Q_3
64	20	20+157=177
65	10	65+177=187
66	8	8+187=195
	N=195	

Q_1 = The size of $\left(\frac{N+1}{4}\right)$ item

= The size of 49^{th} item.

$\boxed{Q_1 = 60}$

Q_3 = The size of $\left[3\left(\frac{N+1}{4}\right)\right]$ item

= The size of 147^{th} item.

$\boxed{Q_3 = 63}$

Quartile deviation $Q.D = \frac{Q_3 - Q_1}{2} = \frac{63-60}{2} = 1.5$

Co-efficient of $Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_3} = \frac{63-60}{63+60} = 0.0244.$

Quartile deviation - continuous series:

Find Q.D and its co-efficien

$$Q.D \text{ and } = \frac{Q_3 - Q_1}{2}$$

$$\text{Co-eff of } Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Where,
$$Q_1 = L + \left[\frac{\frac{N}{4} - c.f}{f}\right] \times c$$

$$Q_3 = L + \left[\frac{\frac{3N}{4} - c.f}{f}\right] \times c$$

1. find Q.D and its co-efficient from the following table.

C.I	4 - 8	8 - 12	12-16	16-20	20-24	24-28	28-32	~~32-3~~
f	6.	10	18	30	15	12	10	
C.I	32 - 36	36 - 40						
f	6	2						

A:-

C.I	f	C.f	
4 - 8	6	6	
8 - 12	10	10+6= 16	C.f for Q_1
L 12 - 16	18 (f)	18+16= 34	
16 - 20	30	34+30 = 64	
20 - 24	15	64+15 = 79	c.f for Q_3
L 24 - 28	12 f	79+12 = 91	
28 - 32	10	91+10 = 101	
32 - 36	6	101 + 6 = 107	
36 - 40	2	107+2 = 109	
	N = 109		

i) $\frac{N}{4} = \frac{109}{4} = 27.25$, $c = 4$, $c.f = 16$, $L = 12$, $f = 18$

$\therefore Q_1 = L + \left(\frac{\frac{N}{4} - c.f}{f}\right) \times c$

$= 12 + \left(\frac{27.25 - 16}{18}\right) \times 4$

$= 12 + (0.6250) \times 4$

$= 12 + 2.5 = 14.5$

$\boxed{Q_1 = 14.5}$

ii) $\frac{3N}{4} = 3(27.25) = 81.75$

$f = 12$, $c.f = 79$, $L = 24$, $c = 4$

$\therefore Q_3 = L + \left(\frac{\frac{3N}{4} - c.f}{f}\right) \times c$

$= 24 + \left(\frac{81.75 - 79}{12}\right) \times 4$

$= 24 + 0.9167$

$\boxed{Q_3 = 24.9167}$

$$Q.D = \frac{Q_3 - Q_1}{2} = \frac{24.9167 - 14.5}{2} = 5.2084$$

$$\text{Co-eff of } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{24.9167 - 14.5}{24.9167 + 14.5} = 0.2643\ \%$$

Merits :

1. It is ~~si~~ easy simple to understand and calculate.
2. It is not influenced by extreme values.
3. It can be find out with open-end classes.

Demerits :

1. It ignores first 25 percent of the items and last 25 percent of the items.
2. It is a positional average, hence it is not useful for further mathematical treatment.
3. It gives only rough answer

Mean deviation (M.D):

Mean deviation is the arithematic mean of the deviations of a series computed from any measure of central tendency. i.e., either mean or median or mode.

Mean deviation- individual series:

$$M.D = \frac{\Sigma |D|}{n}$$, where n = no. of observations.

$$|D| = \left| x - \begin{pmatrix} \text{Mean} \\ \text{(or)} \\ \text{Median} \\ \text{(or)} \\ \text{Mode} \end{pmatrix} \right|$$

$$\text{Co-efficient of } M.D = \frac{M.D}{\text{Mean (or) Median (or) Mode}}$$

1. Calculate M.D from Mean and Mediam from the following values.

100, 200, 150, 360, 250, 490, 600, 500, 671.

Ans: i) From mean:

$$\bar{x} = \frac{\Sigma x_i}{n} = \frac{1}{9}(100+200+150+360+250+490+600+500+671)$$

$$\bar{x} = \frac{3321}{9}$$

$$\bar{x} = 369$$

$$\therefore |D| = |x - \bar{x}| = |x - 369|$$

x	$\lvert D\rvert = \lvert x-\bar{x}\rvert$
100	269
200	169
150	219
360	9
250	119
490	121
600	231
500	131
671	302
	$\Sigma\lvert D\rvert = 1570$

$$M.D = \frac{\Sigma |D|}{n} = \frac{1570}{9} = 174.4444.$$

(ii) From median:

Ascending order is 100, ~~110~~ 150, 200, 250, 360, 490, 500, 600, 671

n = no. of observations = 9 (odd)

Median = value of $\left(\frac{n+1}{2}\right)$ item

$= \left(\frac{9+1}{2}\right) = 5$ item $= 360$

$\therefore$ Median = 360.

$\therefore |D| = |x - \text{Median}| = |x - 360|$

x	$\lvert D \rvert = \lvert x - 360 \rvert$
100	260
200	160
150	210
360	0
250	110
490	130
600	240
500	140
671	311
	$\Sigma \lvert D \rvert = 1561$

$M.D = \frac{\Sigma |D|}{n} = \frac{1561}{9} = 173.4444$

co-efficient of $M.D = \frac{M.D}{\text{Mean}} = \frac{174.444}{369} = 0.4727$.

co-eff of $M.D = \frac{M.D}{\text{Median}} = \frac{173.4444}{360} = 0.4818$.

Mean deviation - Discrete series and continuous series:

$$M.D = \frac{\Sigma f \cdot |D|}{N}$$

where, $N = \Sigma f$

$$|D| = \left| x - \begin{pmatrix} \text{Mean} \\ \text{(or)} \\ \text{Median} \\ \text{(or)} \\ \text{Mode} \end{pmatrix} \right|$$

$$\text{co-eff of } M.D = \frac{M.D}{\text{Mean (or) Median (or) Mode}}.$$

1. Calculate M.D from (i) Mean, (ii) Median. And also find its co-efficient from the following data.

x :	10	15	20	30	40	50
f :	8	12	15	10	3	2

Mean → M.D = 7.84

Median → M.D = 7.2

2. Calculate M.D from the Mode to the following data.

M.D = 14.3333

C.I :	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
f :	5	8	7	12	28	20	10	10

Standard deviation (S.D):

The standard deviation is the square root of the average of the squares of deviation from mean. And it is denoted by σ (sigma).

i.e., $\sigma = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n}}$

Standard deviation - individual series:

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

where, n = No: of observations.

Standard deviation - Discrete series:

$$\sigma = \sqrt{\frac{\Sigma f x^2}{N} - \left(\frac{\Sigma f x}{N}\right)^2}, \text{ where } N = \Sigma f$$

(or)

$$\sigma = \left(\sqrt{\frac{\Sigma f d^2}{N} - \left(\frac{\Sigma f d}{N}\right)^2}\right) \times C$$

1. Calculate standard deviation to the following values

10, 12, 14, 16, 18, 20, 22, 24, 26

A:-

x	x^2
10	100
12	144
14	196
16	256
18	324
20	400
22	484
24	576
26	676
$\Sigma x = 162$	$\Sigma x^2 = 3156$

$$S.D = \sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$= \sqrt{\frac{3156}{9} - \left(\frac{162}{9}\right)^2} = 5.1640$$

1. calculate standard deviation from the following data.

x :	4	5	6	7	8	9	10
f :	6	8	10	15	12	10	9

A:-

x	f	fx	x^2	fx^2
4	6	24	16	96
5	8	40	25	200
6	10	60	36	360
7	15	105	49	735
8	12	96	64	768
9	10	90	81	810
10	9	90	100	900
	$N=70$	$\Sigma fx=505$		$\Sigma fx^2=3869$

$$S.D = \sigma = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\frac{3869}{70} - \left(\frac{505}{70}\right)^2} = 1.7960.$$

Standard deviation- continuous series:

$$\sigma = \sqrt{\frac{\Sigma f x^2}{N} - \left(\frac{\Sigma f x}{N}\right)^2}$$, where $N = \Sigma f$

x = Middle value of each class.

1. Calculate S.D to the following frequency distribution

C.I :	30-32	32-34	34-36	36-38	38-40	40-42	42-44
f :	2	9	25	30	49	62	39

C.I :	44-46	46-48	48-50
f :	20	11	3

Coefficient of variation (C.V):

The standard deviation is an absolute measure of dispersion. The relative measure is known as co-efficient of variation. It is used to compare the variability of two or more groups.

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

where, σ = standard deviation.

1. The prices of ~~strai~~ shares x and y are given below, which is more consistant or stable.

x : 55 54 52 53 56 58 52 50 51 49

y : 108 107 105 105 106 107 107 103 104 101

A:

x		y	
x	x^2	y	y^2
55	3025	108	11664
54	2916	107	11449
52	2704	105	11025
53	2809	105	11025
56	3136	106	11236
58	3364	107	11449
52	2704	107	11449
50	2500	103	10609
51	2601	104	10816
49	2401	101	10201
$\Sigma x = 530$	$\Sigma x^2 = 28160$	$\Sigma y = 1053$	$\Sigma y^2 = 110923$

Given, n = num of observations = 10

$$\bar{x} = \frac{\Sigma x}{n} = \frac{530}{10} = 53$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{1053}{10} = 105.3$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$ < Individual Discrete series

$$= \sqrt{\frac{28160}{10} - (53)^2}$$

$$= \sqrt{7}$$

$$\boxed{\sigma = 2.6458}$$

$$C.V_{(x)} = \frac{\sigma}{\bar{x}} \times 100$$

$$= \frac{2.6458}{53} \times 100$$

$$= 4.9921$$

$$\sigma = \sqrt{\frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2}$$

$$= \sqrt{\frac{110923}{10} - (105.3)^2}$$

$$= \sqrt{4.21}$$

$$\boxed{\sigma = 2.0518}$$

$$C.V_{(y)} = \frac{\sigma}{\bar{y}} \times 100$$

$$= \frac{2.0518}{105.3} \times 100$$

$$= 1.9485$$

Hence the co-efficient of variation for y series is lessthan x series. So, y series is more consistant than x series.